ECEN 248 - Lab Report

Lab Number: 4

Lab Title: Simple Arithmetic Logic

Section Number: 509

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Date: 10/5/2022

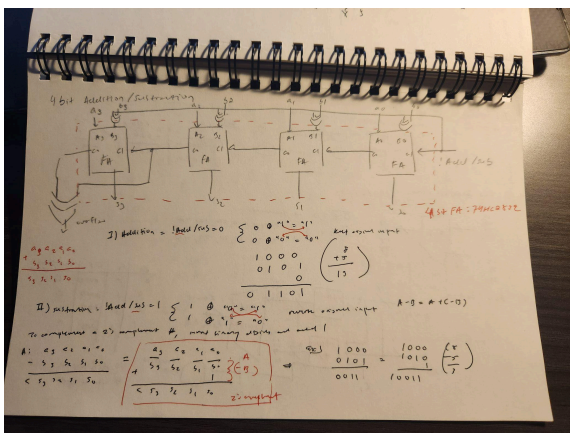
TA: Sri Kodi

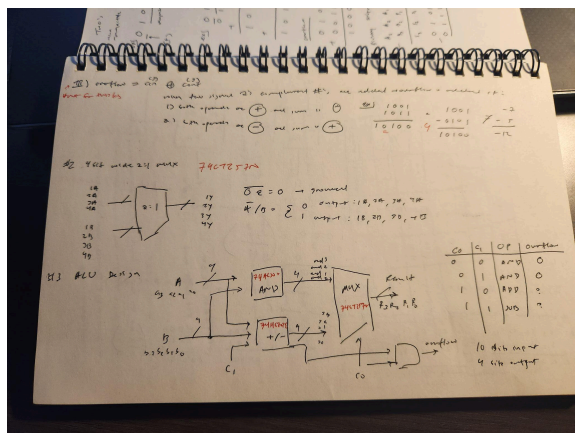
Objectives:

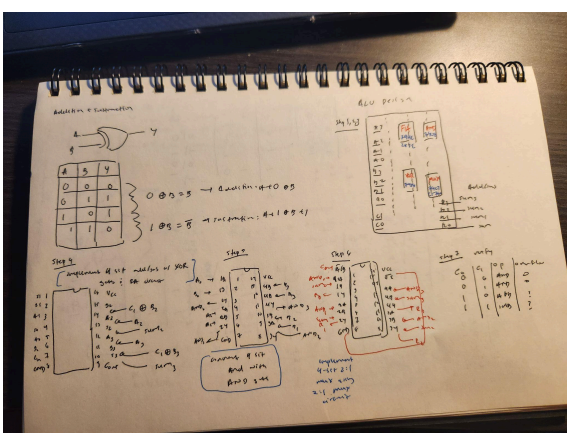
The purpose of this lab is to create a simple arithmetic logic unit. The ALU will perform computations like addition and subtraction and bit-wise AND. The concept of two's complements was introduced as well as multiplexers. Two's complement allowed for negative numbers to be considered.

Design:

To complete this lab, a design of a 4-bit ALU was created. A 4-bit addition/subtraction was created using a 4-bit carry ripple adder (with the component SN74HC283E) and an XOR gate. The circuit was then tested with three LEDS to see if it worked before implementing the full ALU design. An AND gate was then connected to the circuit. After that, a 2:1 MUX was introduced to the circuit using a SN74CT257N chip. The AND outputs and addition/subtraction inputs were connected to the MUX. Making sure OE was connected to the ground. This completed the ALU design which was then tested using three LEDS.



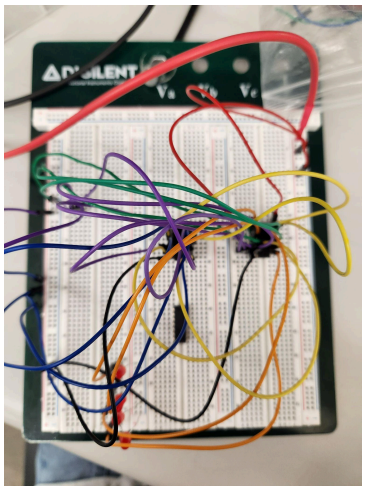


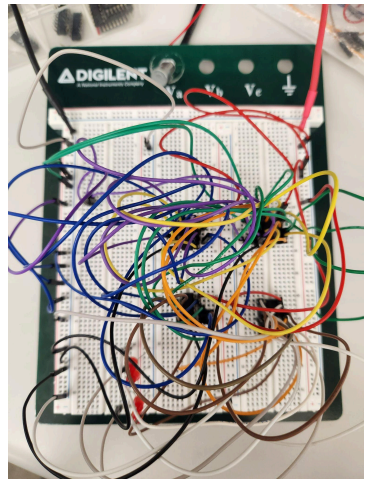


Designing ALU

Results:

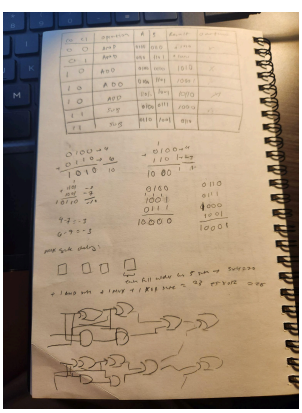
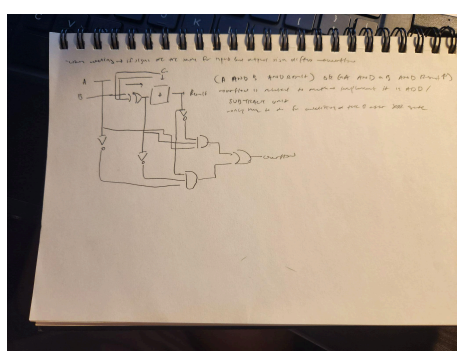
The implementation of the ALU went as planned. The addition and subtraction unit worked as well with three LED outputs.





Addition/Subtraction unit and full ALU

Conclusion:

In this lab, a simple ALU was created starting from a addition/subtraction unit comprised of a 4-bit carry ripple adder and XOR gate. A MUX and AND gate were then added to the circuit in order to complete the ALU design. From this lab, two's complement was explored as well as how to use a MUX and a 4-bit carry ripple adder without having to implement separate logic gates (full adder circuits).

Post-lab Deliverables: